Classical Chinese Mathematics

孫子算經

Sunzi Suanjing

Master Sun's Mathematical Manual

Attributed to **Sunzi** (孫子, Master Sun)

c. 3rd–5th century CE

One of the Ten Computational Canons (算經十書)
Listed during the Tang dynasty (618–907 CE)
Contains the earliest known description of the Chinese Remainder Theorem

Complete Edition with full original Chinese text
Preface (序) · Volume I (卷上) · Volume II (卷中) · Volume III (卷下)
Including the famous 「物不知數」 problem (CRT)

May 28, 2026

1 Introduction

The *Sunzi Suanjing* (孫子算經, "Master Sun's Mathematical Manual") is one of the earliest surviving Chinese mathematical texts, written during the 3rd to 5th centuries CE. It was listed as one of the *Ten Computational Canons* (算經十書) during the Tang dynasty. The identity of its author, Sunzi ("Master Sun"), remains unknown, though he lived much later than his namesake Sun Tzu, the author of *The Art of War*.

The treatise consists of three volumes (卷):

- **Volume I** (卷上): Units of measurement, counting rod notation, arithmetic algorithms, and the multiplication table from 9×9 to 1×1
- **Volume II** (卷中): Fractions, geometry (areas, volumes), and practical problems including equal distribution and proportional division
- **Volume III** (卷下): Practical problems including the famous "Problem of the Unknown Quantity"—the earliest known statement of what is now called the **Chinese Remainder Theorem**

Historical Significance

The *Sunzi Suanjing* holds a special place in the history of mathematics. While it covers routine arithmetic, fractions, and practical calculations, its most celebrated contribution is **Problem 26 of Volume III**:

> 今有物，不知其數。三三數之賸二，五五數之賸三，七七數之賸二。問物幾何?
>
> *"There are certain things whose number is unknown. If we count them by threes, we have two left over; by fives, we have three left over; and by sevens, two are left over. How many things are there?"*

This problem, known as the "Problem of the Unknown Quantity" (物不知數), represents the earliest known statement of the Chinese Remainder Theorem. The method described in the text provides the solution:

$$N = 70 \times r_1 + 21 \times r_2 + 15 \times r_3 - 105 \times k$$

where r_1, r_2, r_3 are the remainders upon division by 3, 5, and 7 respectively.

The general theorem was later fully developed by Qin Jiushao (秦九韶) in his *Mathematical Treatise in Nine Sections* (數書九章, 1247 CE), where he called it the "Great Extension Seeking-Unity Method" (大衍求一術).

Preface (序)

孫子曰：夫算者，天地之經緯，群生之元首；五常之本末，陰陽之父母；星辰之建號，三光之表裹；五行之準平，四時之終始；萬物之祖宗，六藝之綱紀。稽群倫之聚散，考二氣之降升；推寒暑之迭運，步遠近之殊同；觀天道精微之兆基，察地理從橫之長短；采神祇之所在，極成敗之符驗；窮道德之理，究性命之情。立規矩，準方圓，謹法度，約尺丈，立權衡，平重輕，剖毫釐，析黍絫；歷億載而不朽，施八極而無疆。散之不可勝究，斂之不盈掌握。嚮之者富有餘，背之者貧且窶；心開者幼沖而即悟，意閉者皓首而難精。夫欲學之者必務量能揆己，志在所專。如是則焉有不成者哉。

Translation:

Sun Zi said: Calculation is the meridian and latitude of heaven and earth, the origin and head of all living beings; the root and branch of the Five Constants, the father and mother of yin and yang; the establishment of titles for stars and constellations, the exterior and interior of the Three Lights; the standard balance of the Five Elements, the beginning and end of the Four Seasons; the ancestors of all things, the framework and discipline of the Six Arts. Investigate the gathering and dispersing of all categories, examine the descending and ascending movements of the Two Qi; track the alternating cycles of cold and heat, pace the differences and similarities between near and far; observe the subtle omens and foundations of the heavenly Dao, examine the lengths and widths of the horizontal and vertical features of geography; gather signs of where gods and spirits reside, thoroughly test omens of success and failure; exhaust the principles of virtue and morality, investigate the nature and essence of life. Establish rules and compasses, standardize squares and circles, observe laws and regulations carefully, measure with precision in *chi* and *zhang*, set up weights and balances, equalize the heavy and light, dissect thousandths of an inch, analyze grains accumulated one by one; enduring through countless generations without decay, its influence extends to all directions without limit. When dispersed, it is beyond full comprehension; when gathered, it does not fill the palm of one's hand. Those who face toward it are rich with abundance, while those who turn away from it are poor and destitute; those whose minds open understand even as children, while those whose intentions remain closed find mastery difficult even with a white beard. Therefore, one who wishes to study must necessarily measure his own ability and assess himself carefully, setting his mind on specialization in a particular field. If so, how could there be any failure?

2 Volume I (卷上)

Volume I establishes the foundations of calculation: units of measurement, counting rod notation, arithmetic algorithms, and the multiplication table. The complete Volume I comprises the following eight sections.

Section 1. The Origin of Measurements (度之所起)

度之所起

度之所起，起於忽。欲知其忽，蠶所生，吐絲為忽。十忽為一秒，十秒為一毫，十毫為一釐，十釐為一分，十分為一寸，十寸為一尺，十尺為一丈，十丈為一引；五十尺為一端；四十尺為一疋；六尺為一步。二百四十步為一畝。三百步為一里。

The origin of measurement begins with the "hu" (忽). To understand the hu, it is said that silkworms produce it by spinning silk. Ten hu make one miao (秒), ten miao make one hao (毫), ten hao make one li (釐), ten li make one fen (分), ten fen make one cun (寸), ten cun make one chi (尺), ten chi make one zhang (丈), and ten zhang make one yin (引); fifty chi make one duan (端); forty chi make one pi (疋); six chi make one bu (步). Two hundred and forty bu make one mu (畝). Three hundred bu make one li (里).

稱之所起

稱之所起，起於黍。十黍為一絫，十絫為一銖，二十四銖為一兩，十六兩為一斤，三十斤為一鈞，四鈞為一石。

The origin of weight measurement begins with the shu grain (黍). Ten shu make one lei (絫), ten lei make one zhu (銖), twenty-four zhu make one liang (兩), sixteen liang make one jin (斤), thirty jin make one gou (鈞), and four gou make one dan (石).

量之所起

量之所起，起於粟。六粟為一圭，十圭為一抄，十抄為一撮，十撮為一勺，十勺為一合，十合為一升，十升為一斗，十斗為一斛。斛得六千萬粟。所以得知者，六粟為一圭，十圭六十粟為一抄，十抄六百粟為一撮，十撮六千粟為一勺，十勺六萬粟為一合，十合六十萬粟為一升，十升六百萬粟為一斗，十斗六千萬粟為一斛。十斛六億粟，百斛六兆粟，千斛六京粟，萬斛六陔粟，十萬斛六秭粟，百萬斛六壤粟，千萬斛六溝粟，萬萬斛為一億斛六澗粟，十億斛六正粟，百億斛六載粟。

The origin of volume measurement begins with the su grain (粟). Six su make one gui (圭), ten gui make one chao (抄), ten chao make one cuo (撮), ten cuo make one shao (勺), ten shao make one ge (合), ten ge make one sheng (升), ten sheng make one dou (斗), and ten dou make one hu (斛). One hu contains sixty million su grains. Continuing: ten hu contain six hundred million su; one hundred hu, six zhao; one thousand hu, six jing; ten thousand hu, six gai; one hundred thousand hu, six zi; one million hu, six rang; ten million hu, six gou; one hundred million hu (one yi hu), six jian; ten yi hu, six zheng; one hundred yi hu, six zai.

Section 2. Large Numbers (凡大數之法)

凡大數之法

凡大數之法，萬萬曰億，萬萬億曰兆，萬萬兆曰京，萬萬京曰陔，萬萬陔曰秭，萬萬秭曰壤，萬萬壤曰溝，萬萬溝曰澗，萬萬澗日正，萬萬正曰載。

Generally, the method for large numbers is as follows: ten thousand times ten thousand is yi (億); ten thousand times yi is zhao (兆); ten thousand times zhao is jing (京); ten thousand times jing is gai (陔); ten thousand times *gai is zi (秭); ten thousand times zi is rang (壤); ten thousand times rang is gou (溝); ten thousand times gou is jian (澗); ten thousand times jian is zheng (正); and ten thousand times zheng is zai (載).*

Section 3. Geometric Ratios (周三徑一)

周三徑一

周三徑一。方五邪七；見邪求方，五之，七而一；見方求邪，七之，五而一。

The circumference of a circle is three times its diameter (周三徑一). A square with side length five has a diagonal of seven; to find the side from the diagonal, multiply by five and divide by seven; to find the diagonal from the side, multiply by seven and divide by five.

黃金方寸

黃金方寸重一斤。白金方寸重一十四兩。玉方寸重一十二兩。銅方寸重七兩半。鉛方寸重九兩半。鐵方寸重六兩。石方寸重三兩。

One cubic cun of gold weighs one jin. One cubic cun of silver weighs fourteen liang. One cubic cun of jade weighs twelve liang. One cubic cun of copper weighs seven and a half liang. One cubic cun of lead weighs nine and a half liang. One cubic cun of iron weighs six liang. One cubic cun of stone weighs three liang.

Section 4. Counting Rod Notation (凡算之法)

凡算之法

凡算之法，先識其位，一從十橫，百立千僵，千十相望，萬百相當。

The method for calculation begins by identifying the positions: one is vertical (一從), ten is horizontal (十橫); a hundred stands upright (百立) and a thousand lies flat (千僵). Ten hundreds face each other (千十相望), and thousands and tens correspond to each other in position (萬百相當).

This describes the traditional Chinese counting rod numeral system:

- Units (1–9) are represented by **vertical** rods
- Tens (10–90) are represented by **horizontal** rods
- Hundreds (100–900) are represented by **vertical** rods
- Thousands (1000–9000) are represented by **horizontal** rods

The pattern alternates between vertical and horizontal for each successive place value.

Section 5. Method of Multiplication (凡乘之法)

凡乘之法

凡乘之法，重置其位。上下相觀，上位有十步至十，有百步至百，有千步至千。以上命下，所得之數列於中位。言十即過，不滿自如。上位乘訖者先去之。下位乘訖者則俱退之。六不積，五不隻。上下相乘，至盡則已。

In multiplication, the positions are arranged again for calculation. By observing vertically and horizontally, if the upper position has ten, it moves to the tens place; if it has a hundred, it moves to the hundreds place; if it has a thousand, it moves to the thousands place. The upper number is used to multiply the lower one, and the *resulting product is placed in the middle position. If it reaches ten, carry over; if less than ten, leave as is. After the upper position has been multiplied, remove it first. Once the lower position is multiplied, both are moved back. Six does not accumulate; five does not count as a single unit. Multiplying vertically and horizontally continues until the end is reached.*

Multiplication Example: 81×81

問

九九八十一，自相乘，得幾何?

答

答曰：六千五百六十一。

術

術曰：重置其位，以上八呼下八，八八六十四，即下六千四百於中位。以上八呼下一，一八如八，即於中位下八十。退下位一等，收上位八十。以上位一呼下八，一八如八，即於中位下八十。以上一呼下一，一一如一，即於中位下一。上下位俱收，中位即得六千五百六十一。

Nine times eighty-one multiplied by itself: Arrange the positions again, call out eight from above to multiply with eight below; eight eights are sixty-four. Place six thousand four hundred in the middle position. Call out eight from above to one below, one eight is eight; place eighty in the middle position. Move the lower position back by one rank. Call out one from above to eight below, one eight is eight, place eighty in the middle position. Call out one from above to one below; one times one is one, so place one in the middle position. After collecting both positions, the middle position yields six thousand five hundred sixty-one.

Section 6. Method of Division (凡除之法)

凡除之法

凡除之法，與乘正異。乘得在中央，除得在上方。假令六為法，百為實。以六除百，當進之二等，令在正百下，以六除一，則法多而實少，不可除，故當退就十位。以法除實，言一六而折百為四十，故可除。若實多法少，自當百之，不當復退。故或步法十者置於十位，百者置於百位。上位有空絕者，法退二位。餘法皆如乘時。實有餘者，以法命之，以法為母。實餘為子。

The method of division differs from multiplication. In multiplication, the result is placed in the center; in division, it appears at the top. Suppose six is the divisor (法) and one hundred is the dividend (實). Dividing one hundred by six, it should be advanced two positions. Dividing one by six results in a larger divisor than dividend, which is not possible; therefore, retreat to the tens position. Using the divisor to divide the dividend: saying one six and reducing a hundred by sixty results in forty. If the dividend is large and the divisor small, it naturally belongs to the hundreds place. If there is an empty space at the upper position, the divisor must retreat two positions. The remaining methods follow those of multiplication. If there is a remainder in the dividend, name it using the divisor as the denominator (母) and the remainder as the numerator (子).

Division Example: $6561 \div 9$

問

六千五百六十一，九人分之，問人得幾何?

答

答曰：七百二十九。

術

術曰：先置六千五百六十一於中位，為實。下列九人為法。上位置七百，以上七呼下九，七九六十三，即除中位六千三百。退下位一等，即上位置二十。以上二呼下九，二九十八，即除中位一百八十。又更退下位一等，即上位更置九，即以上九呼下九，九九八十一，即除中位八十一。中位盡，收下位。上位所得即人之所得。自八八六十四至一一如一，並准此。

Six thousand five hundred sixty-one is to be divided among nine people. Place 6561 in the middle position as the dividend (實). Place nine people below as the divisor (法). Place seven hundred in the upper position; call out seven from above to multiply by nine below, seven nines are sixty-three, subtract six thousand three hundred from the middle. Move the lower position back one rank; place twenty in the upper position. Call out two from above to multiply by nine from below, two nines are eighteen. Move the lower position back another rank, place nine in the upper position. Call out nine from above to multiply by nine below; nine nines are eighty-one. When the middle position is exhausted, collect the lower positions. The result in the upper position is what each person receives. From eight eights being sixty-four to one times one being one, follow this method accordingly.

Section 7. Grain Conversion Rates (以粟求糲米)

以粟求糲米

以粟求糲米，三之，五而一。以糲米求粟，五之，三而一。以糲米求飯，五之，二而一。以粟米求糲飯，六之，四而一。以糲飯求糲米，二之，五而一。以綹米求飯，八之，四而一。

To convert su grain to millet rice (糲米), multiply by three and divide by five. To convert millet rice back to su grain, multiply by five and divide by three. To convert millet rice into cooked rice (飯), multiply by five and divide by two. To convert su grain into cooked rice, multiply by six and divide by four. To convert cooked rice back to millet rice, multiply by two and divide by five.

十分減一者

十分減一者，以二乘，二十除。減二者，以四乘，二十除。減三者，以六乘，二十除。減四者，以八乘，二十除。減五者，以十乘，二十除。減六者，以十二乘，二十除。減七者，以十四乘，二十除。減八者，以十六乘，二十除。減九者，以十八乘，二十除。

A reduction of one-tenth is calculated by multiplying by two and dividing by twenty. A reduction of two-tenths by multiplying by four and dividing by twenty, continuing through a reduction of nine-tenths by multiplying by eighteen and dividing by twenty.

Section 8. The Multiplication Table (九九八十一)

Volume I concludes with the complete “nine-nines” multiplication table, recited in reverse order from $9 \times 9 = 81$ down to $1 \times 1 = 1$:

九九八十一，八九七十二，七九六十三，六九五十四，五九四十五，四九三十六，三九二十七，二九一十八，一九如九。
八八六十四，七八五十六，六八四十八，五八四十，四八三十二，三八二十四，二八十六，一八如八。
七七四十九，六七四十二，五七三十五，四七二十八，三七二十一，二七一十四，一七如七。
六六三十六，五六三十，四六二十四，三六一十八，二六十二，一六如六。
五五二十五，四五二十，三五十五，二五一十，一五如五。
四四一十六，三四一十二，二四如八，一四如四。
三三九，二三如六，一三如三。
二二如四，一二如二。
一一如一。

The complete nine-nines song: Nine nines are eighty-one, eight nines are seventy-two, seven nines are sixty-three, six nines are fifty-four, five nines are forty-five, four nines are thirty-six, three nines are twenty-seven, two nines are eighteen, one nine is as nine. And so on down to: one times one is as one.

3 Volume II (卷中)

Volume II contains 28 problems dealing with fractions, areas, volumes, proportional distribution, and practical calculations.

Problem 1. Reducing Fractions

問

今有一十八分之一十二。問約之得幾何?

答

答曰：三分之二。

術

術曰：置十八分在下，一十二分在上。副置二位，以少減多，等數得六，為法。約之，即得。

Now there is twelve eighteenths. Ask: reducing it, what does it equal? Answer: Two thirds. Method: Place eighteen parts below and twelve parts above. Set up both positions, subtract the smaller from the larger; the equal number obtained is six, which is the divisor (法). Reduce by it.

Problem 2. Adding Fractions

問

今有三分之一，五分之二。問合之得幾何?

答

答曰：一十五分之一十一。

術

術曰：置三分、五分在右方，之一、之二在左方。母互乘子，五分之二得六，三分之一得五。并之，得一十一，為實。右方二母相乘，得一十五，為法。不滿法，以法命之，即得。

Now there is one-third and two-fifths. Ask: when combined, what does it equal? Answer: Eleven fifteenths. Method: Place three parts and five parts on the right side; one and two on the left side. Cross-multiply numerators and denominators: two-fifths becomes six, one-third becomes five. Add them to get eleven as the dividend (實). Multiply the two denominators to get fifteen as the divisor (法).

Problem 3. Subtracting Fractions

問

今有九分之八，減其五分之一。問餘幾何?

答

答曰：四十五分之三十一。

術

術曰：置九分、五分在右方，之八、之一在左方。母互乘子，五分之一得九，九分之八得四十。以少減多，餘三十一，為實。母相乘得四十五，為法。不滿法，以法命之，即得。

Problem 4. Averaging Fractions

問

今有三分之一，三分之二，四分之三。問減多益少，幾何而平？

答

答曰：減四分之三者二，減三分之二者一，并以益三分之一，而各平於一十二分之七。

術

術曰：置三分、三分、四分在右方，之一、之二、之三在左方。母互乘子，副并得六十三，置右，為平實。母相乘，得三十六，為法。以列數三乘未并者及法。等數得九，約訖。減四分之三者二，減三分之二者一，并以益三分之一，各平於一十二分之七。

Now there are one-third, two-thirds, and three-fourths. Ask: taking from the greater and adding to the lesser, how much is needed to equalize? Answer: Subtract two from three-fourths, subtract one from two-thirds, and add both to one-third, so that each becomes equal to seven twelfths.

Problem 5. Grain Conversion — Glutinous Rice

問

今有粟一斗，問為糯米幾何？

答

答曰：六升。

術

術曰：置粟一斗，十升。以糲米率三十乘之，得三百升，為實。以粟率五十為法，除之，即得。

Problem 6. Grain Conversion — Barnyard Millet

問

今有粟二斗一升，問為稗米幾何？

答

答曰：一斗一升五十分升之一十七。

術

術曰：置粟二十一升。以稗米率二十七乘之，得五百六十七升，為實。以粟率五十為法，除之。不盡，以法命分。

Problem 7. Grain Conversion — Pan Rice

問

今有粟四斗五升，問為槃米幾何？

答

答曰：二斗一升五分升之三。

術

術曰：置粟四十五升。以二約槃米率二十四，得一十二。乘之，得五百四十升，為實。以二約粟率五十，得二十五，為法。除之。不盡，以等數約之，而命分。

Problem 8. Grain Conversion — Imperial Rice

問

今有粟七斗九升，問為御米幾何？

答

答曰：三斗三升一合八勺。

術

術曰：置七斗九升。以御米率二十一乘之，得一千六百五十九升，為實。以粟率五十除之，即得。

Problem 9. Brick Calculation

問

今有屋基南北三丈，東西六丈，欲以磚砌之。凡積二尺，用磚五枚。問計幾何？

答

答曰：四千五百枚。

術

術曰：置東西六丈，以南北三丈乘之，得一千八百尺。以五乘之，得九千尺。以二除之，即得。

Problem 10. Round Cellar Volume

問

今有圓窖下周二百八十六尺，深三丈六尺。問受粟幾何？

答

答曰：一十五萬一千四百七十四斛七升二十七分升之一十一。

術

術曰：置周二百八十六尺，自相乘，得八萬一千七百九十六尺。以深三丈六尺乘之，得二百九十四萬四千六百五十六尺。以一十二除之，得二十四萬五千三百八十八尺。以斛法一尺六寸二分除之，即得。

Problem 11. Square Cellar Volume

問

今有方窖廣四丈六尺，長五丈四尺，深三丈五尺。問受粟幾何?

答

答曰：五萬三千六百六十六斛六斗六升三分升之二。

術

術曰：置廣四丈六尺，長五丈四尺，相乘，得二千四百八十四尺。以深三丈五尺乘之，得八萬六千九百四十尺。以斛法一尺六寸二分除之，即得。

Problem 12. Circular Cellar Volume

問

今有圓窖周五丈四尺，深一丈八尺。問受粟幾何?

答

答曰：二千七百斛。

術

術曰：先置周五丈四尺，自相乘得二千九百一十六尺。以深一丈八尺乘之，得五萬二千四百八十八尺。以一十二除之，得四千三百七十四尺。以斛法一尺六寸二分除之，即得。

Problem 13. Area of a Circular Field

問

今有圓田周三百步，徑一百步。問得田幾何?

答

答曰：三十一畝奇六十步。

術

術曰：先置周三百步，半之，得一百五十步。又置徑一百步，半之，得五十步。相乘，得七千五百步。以畝法二百四十步除之，即得。

Alternative methods given:

又術

又術：周自相乘，得九萬步。以一十二除之，得七千五百步。以畝法除之，得畝數。
又術：徑自乘，得一萬。以三乘之，得三萬步。四除之，得七千五百步。以畝法除之，得畝數。

Problem 14. Square Field with Mulberry Tree

問

今有方田，桑生中央。從角至桑一百四十七步。問為田幾何?

答

答曰：一頃八十三畝奇一百八十步。

術

術曰：置角至桑一百四十七步，倍之，得二百九十四步。以五乘之，得一千四百七十步。以七除之，得二百一十步。自相乘，得四萬四千一百步。以二百四十步除之，即得。

Problem 15. Wooden Pillows from a Block

問

今有木方三尺，高三尺。欲方五寸作枕一枚，問得幾何?

答

答曰：二百一十六枚。

術

術曰：置方三尺，自相乘，得九尺。以高三尺乘之，得二十七尺。以一尺木八枕乘之，即得。

Problem 16. Volume of a Dike

問

今有堤，下廣五丈，上廣三丈，高二丈，長六十尺。欲以一千尺為一方，問計幾何?

答

答曰：四十八方。

術

術曰：置堤上廣三丈，下廣五丈。并之，得八丈。半之，得四丈。以高二丈乘之，得八百尺。以長六十尺乘之，得四萬八千。以一千尺除之，即得。

Problem 17. Volume of a Ditch

問

今有溝廣十丈，深五丈，長二十丈。欲以千尺作一方，問得幾何?

答

答曰：一千方。

術

術曰：置廣一十丈，以深五丈乘之，得五千尺。又以長二十丈乘之，得一百萬尺。以一千除之，即得。

Problem 18. Square Root Extraction

問

今有積二十三萬四千五百六十七步。問為方幾何?

答

答曰：四百八十四步九百六十八分步之三百一十一。

術

術曰：置積二十三萬四千五百六十七步，為實。次借一算為下法。步之，超一位，至百而止。商置四百。於實之上，副置四萬於實之下，下法之上，名為方法。命上商四百，除實。除訖，倍方法，方法一退，下法再退。復置上商八十，以次前商。副置八百於方法之下，下法之上，名為廉法。命上商八十，以方法、廉法各除實。訖，倍廉法，從方法。方法一退，下法再退。復置上商四，以次前。副置四於方法之下，下法之上，名曰隅法。命上商四，以方法、廉法、隅法各除實。訖，倍隅法，從方法。上商得四百八十四，下法得九百六十八，不盡三百一十一。是為方四百八十四步九百六十八分步之三百一十一。

This is the method of square root extraction (開平方), one of the most important algorithms in Chinese mathematics.

Problem 19. Circle from Area

問

今有積三萬五千步。問為圓幾何?

答

答曰：六百四十八步一千二百九十六分步之九十六。

術

術曰：置積三萬五千步，以十二乘之，得四十二萬步，為實。次借一算為下法。步之，超一位，至百而止。商置六百。於實之上，副置六萬於實之下，下法之上，名為方法。命上商六百，除實。除訖，倍方法，方法一退，下法再退。復置上商四十，以次前商。副置四百於方法之下，下法之上，名為廉法。命上商四十，以方法、廉法各除實。訖，倍廉法，從方法。方法一退，下法再退。復置上商八，以次前商。副置八於方法之下，下法之上，名曰隅法。命上商八，以方法、廉法、隅法各除實。訖，倍隅法，從方法。上商得六百四十八，下法得一千二百九十六，不盡九十六。是為圓徑六百四十八步一千二百九十六分步之九十六。

Problem 20. Area of a Hill Field

問

今有丘田周六百三十九步，徑三百八十步。問為田幾何?

答

答曰：二頃五十二畝二百二十五步。

術

術曰：半周得三百一十九步五分，半徑得一百九十步，二位相乘，得六萬七百五步。以畝法除之，即得。

Problem 21. City Wall Construction

問

今有築城，上廣二丈，下廣五丈四尺，高三丈八尺，長五千五百五十尺。秋程人功三百尺。問須功幾何?

答

答曰：二萬六千一十一功。

術

術曰：并上、下廣，得七十四尺，半之，得三十七尺。以高乘之，得一千四百六尺。又以長乘之，得積七百八十萬三千三百尺。以秋程人功三百尺除之，即得。

Problem 22. Canal Construction

問

今有穿渠，長二十九里一百四步，上廣一丈二尺六寸，下廣八尺，深一丈八尺。秋程人功三百尺。問須功幾何?

答

答曰：三萬二千六百四十五人，不盡六十九尺六寸。

術

術曰：置里數，以三百步乘之，內零步，六之，得五萬二千八百二十四尺。并上、下廣，得二丈六寸。半之，以深乘之，得一百八十五尺四寸。以長乘，得九百七十九萬三千五百六十九尺六寸。以人功三百尺除之，即得。

Problem 23. Proportional Money Distribution

問

今有錢六千九百三十，欲令八十一人、七十二人、六十三人分之，問人得幾何?

答

答曰：二分，人得錢二十二。三分，人得錢三十三。四分，人得錢四十四。

術

術曰：先置八十一人於上，七十二人次之，六十三人在下。上位以二乘之，得一百六十二；次位以三乘之，得二百一十六；下位以四乘之，得二百五十二。副并三位，得六百三十，為法。又置錢六千九百三十為三位。上位以一百六十二乘之，得一百一十二萬二千六百六十；又以二百一十六乘中位，得一百四十九萬六千八百八十；又以二百五十二乘下位，得一百七十四萬六千三百六十；各為實。以法六百三十各除之，上位得一千七百八十二，中位得二千三百七十六，下位得二千七百七十二。各以人數除之，即得。

Problem 24. Five Feudal Lords and Tangerines

問

今有五等諸侯，共分橘子六十顆。人別加三顆。問五人各得幾何?

答

答曰：公一十八顆。侯一十五顆。伯一十二顆。子九顆。男六顆。

術

術曰：先置人數，別加三顆於下，次六顆，次九顆，次一十二顆，上十五顆。副并之，得四十五。以減六十顆，餘，人數除之，人得三顆。各加不并者，上得一十八，為公分；次得一十五，為侯分；次得十二，為伯分；次得九，為子分；下得六，為男分。

Problem 25. Three People Holding Money

問

今有甲、乙、丙三人持錢。甲語乙、丙:「各將公等所持錢半以益我錢，成九十。」乙復語甲、丙:「各將公等所持錢半以益我錢，成七十。」丙復語甲、乙：「各將公等所持錢半以益我錢，成五十六。」問三人元持錢各幾何?

答

答曰：甲七十二。乙三十二。丙四。

術

術曰：先置三人所語為位，以三乘之，各為積，甲得二百七十，乙得二百一十，丙得一百六十八。各半之，甲得一百三十五，乙得一百五，丙得八十四。又置甲九十、乙七十、丙五十六，各半之。以甲、乙減丙，以甲、丙減乙，以乙、丙減甲，即各得元數。

Problem 26. Woman Weaving — Geometric Progression

問

今有女子善織，日自倍。五日織通五尺。問日織幾何？

答

答曰：初日織一寸三十一分寸之一十九。次日織三寸三十一分寸之七。次日織六寸三十一分寸之一十四。次日織一尺二寸三十一分寸之二十八。次日織二尺五寸三十一分寸之二十五。

術

術曰：各置列衰，副并，得三十一，為法。以五尺乘未并者，各自為實。實如法而一，即得。

Problem 27. Thieves and Silk — Surplus and Deficit

問

今有人盜庫絹，不知所失幾何。但聞草中分絹，人得六匹，盈六匹；人得七匹，不足七匹。問人、絹各幾何？

答

答曰：賊一十三人。絹八十四匹。

術

術曰：先置人得六匹於右上，盈六匹於右下。後置人得七匹於左上，不足七匹於左下。維乘之，所得，并之，為實。并盈不足，為法。除之，即得賊人。以六人乘之，加盈六匹，即得絹數。

This is the "surplus and deficit" (盈不足) method, one of the most important techniques in ancient Chinese mathematics for solving practical problems.

Problem 28. Rope and Wood

問

今有木，不知長短。引繩度之，餘繩四尺五寸。屈繩量之，不足一尺。問木長幾何？

答

答曰：六尺五寸。

術

術曰：置餘繩四尺五寸，加不足一尺，共五尺五寸。倍之，得一丈一尺。減餘四尺五寸，即得。

4 Volume III (卷下)

Volume III contains practical problems covering a wide range of applications, from grain distribution and land measurement to the famous "Problem of the Unknown Quantity" that contains the earliest known statement of the Chinese Remainder Theorem.

Problem 1. Rent Distribution Among Nine Families

問

今有甲、乙、丙、丁、戊、己、庚、辛、壬九家共輸租。甲出三十五斛，乙出四十六斛，丙出五十七斛，丁出六十八斛，戊出七十九斛，己出八十斛，庚出一百斛，辛出二百一十斛，壬出三百二十五斛。凡九家共輸租一千斛。僦運直折二百斛外，問家各幾何?

答

答曰：甲二十八斛。乙三十六斛八蚪。丙四十五斛六蚪。丁五十四斛四蚪。戊六十三斛二蚪。己六十四斛。庚八十斛。辛一百六十八斛。壬二百六十斛。

術

術曰：置甲出三十五斛，以四乘之，得一百四十斛。以五除之，得二十八斛。乙出四十六斛，以四乘之，得一百八十四斛。以五除之，得三十六斛八蚪。丙出五十七斛，以四乘之，得二百二十八斛。以五除之，得四十五斛六蚪。丁出六十八斛，以四乘之，得二百七十二斛。以五除之，得五十四斛四蚪。戊出七十九斛，以四乘之，得三百一十六斛。以五除之，得六十三斛二蚪。己出八十斛，以四乘之，得三百二十斛。以五除之，得六十四斛。庚出一百斛，以四乘之，得四百斛。以五除之，得八十斛。辛出二百一十斛，以四乘之，得八百四十斛。以五除之，得一百六十八斛。壬出三百二十五斛，以四乘之，得一千三百斛。以五除之，得二百六十斛。

Problem 2. Conscription Ratio

問

今有丁一千五百萬，出兵四十萬。問幾丁科一兵?

答

答曰：三十七丁五分。

術

術曰：置丁一千五百萬，為實。以兵四十萬為法。實如法，即得。

Problem 3. Pile of Millet

問

今有平地聚粟，下周三丈六尺，高四尺五寸。問粟幾何?

答

答曰：一百斛。

術

術曰：置周三丈六尺，自相乘，得一千二百九十六尺。以高四尺五寸乘之，得五千八百三十二尺。以三十六除之，得一百六十二尺。以斛法一尺六寸二分除之，即得。

Problem 4. Silk Distribution

問

今有絹七萬八千七百三十二匹，令一百六十二人分之。問人得幾何?

答

答曰：四百八十六匹。

術

術曰：置絹七萬八千七百三十二匹，為實。以一百六十二人為法。實如法，即得。

Problem 5. Go Board

問

今有碁局方一十九道。問用碁幾何?

答

答曰：三百六十一。

術

術曰：置一十九道，自相乘之，即得。

This gives $19^2 = 361$ *intersection points on a standard Go board (碁局).*

Problem 6. Household Cotton Contribution

問

今有三萬六千四百五十四戶，戶輸綿二斤八兩。問計幾何?

答

答曰：九萬一千一百三十五斤。

術

術曰：置三萬六千四百五十四戶，上十之，得三十六萬四千五百四十。以四乘之，得一百四十五萬八千一百六十兩。以十六除之，即得。

Problem 7. Cotton Distribution per Household

問

今有綿九萬一千一百三十五斤，給與三萬六千四百五十四戶。問戶得幾何?

答

答曰：二斤八兩。

術

術曰：置九萬一千一百三十五斤，為實。以三萬六千四百五十四戶為法。除之，得二斤，不盡一萬八千二百二十七斤。以一十六乘之，得二十九萬一千六百三十二兩。以戶除之，即得。

Problem 8. Millet Exchange for Beans

問

今有粟三千九百九十九斛九蚪六升，凡粟九蚪易豆一斛。問計豆幾何?

答

答曰：四千四百四十四斛四蚪。

術

術曰：置粟三千九百九十九斛九斗六升，為實。以九斗為法。實如法，即得。

Problem 9. Additional Millet per Hu

問

今有粟二千三百七十四斛，斛加三升。問共粟幾何?

答

答曰：二千四百四十五斛二斗二升。

術

術曰：置粟二千三百七十四斛，以一斛三升乘之，即得。

Problem 10. Granary Loss Over Nine Years

問

今有粟三十六萬九千九百八十斛七斗，在倉九年，年斛耗三升。問一年、九年各耗幾何?

答

答曰：一年耗一萬一千九十九斛四斗二升一合。九年耗九萬九千八百九十四斛七斗八升九合。

術

術曰：置三十六萬九千九百八十斛七斗，以三升乘之，得一年之耗。又以九乘之，即九年之耗。

Problem 11. Interest on Silk Loan

問

今有貸與人絲五十七斤，限歲出息一十六斤。問斤息幾何?

答

答曰：四兩五十七分兩之二十八。

術

術曰：列限息絲一十六斤，以一十六兩乘之，得二百五十六兩。以貸絲五十七斤除之。不盡，約之，即得。

Problem 12. Buying Silk with Millet

問

今有粟一十二萬八千九百四十斛九斗三合，出與人買絹，一匹直粟三斛五斗七升。問絹幾何?

答

答曰：三萬六千一百一十七匹三丈六尺。

術

術曰：置粟一十二萬八千九百四十斛九斗三合，為實。以三斛五斗七升為法。除之，得匹。餘四十之，所得，又以法除之，即得。

Problem 13. Gold Price per Liang

問

今有黃金一斤，直錢一十萬。問兩直幾何?

答

答曰：六千二百五十錢。

術

術曰：置錢一十萬，以一十六兩除之，即得。

Problem 14. Brocade Price by Length

問

今有錦一疋，直錢一萬八千。問丈、尺、寸各直幾何?

答

答曰：丈，四千五百錢。尺，四百五十錢。寸，四十五錢。

術

術曰：置錢一萬八千，以四除之，得丈之直。一退、再退，得尺、寸之直。

Problem 15. Birds on a Field

問

今有地長一千步，廣五百步。尺有鶉，寸有鷃。問鶉、鷃各幾何?

答

答曰：鶉一千八百萬。鷃一億八千萬。

術

術曰：置長一千步，以廣五百步乘之，得五十萬步。以三十六乘之，得一千八百萬尺，即得鶉數。上十之，即得鷃數。

Problem 16. Three Classes of People Eating

問

今有六萬口，上口三萬人，日食九升；中口二萬人，日食七升；下口一萬人，日食五升。問上、中、下口共食幾何?

答

答曰：四千六百斛。

術

術曰：各置口數，以日食之數乘之。所得，并之，即得。

Problem 17. Bundle of Square Objects

問

今有方物一束，外周一匝有三十二枚。問積幾何?

答

答曰：八十一枚。

術

術曰：重置二位。上位減八，餘加下位。至盡虛加一，即得。

Problem 18. Pole and Shadow

問

今有竿不知長短，度其影得一丈五尺。別立一表，長一尺五寸，影得五寸。問竿長幾何?

答

答曰：四丈五尺。

術

術曰：置竿影一丈五尺，以表長一尺五寸乘之，上十之，得二十二丈五尺。以表影五寸除之，即得。

This is an early example of proportional reasoning using similar triangles (重差).

Problem 19. Rice in a Vessel

問

今有器中米，不知其數。前人取半，中人三分取一，後人四分取一，餘米一斗五升。問本米幾何?

答

答曰：六斗。

術

術曰：置餘米一斗五升，以六乘之，得九斗。以二除之，得四斗五升。以四乘之，得一斛八斗。以三除之，即得。

Problem 20. People and Carts

問

今有三人共車，二車空；二人共車，九人步。問人與車各幾何?

答

答曰：一十五車。三十九人。

術

術曰：置二車，以三乘之，得六。加步者九人，得車一十五。欲知人者，以二乘車，加九人，即得。

Problem 21. Guests at a Feast

問

今有婦人河上蕩桮。津吏問曰:「桮何以多?」婦人曰:「家有客。」津吏曰:「客幾何?」婦人曰:「二人共飯，三人共羹，四人共肉，凡用桮六十五。不知客幾何?」

答

答曰：六十人。

術

術曰：置六十五桮，以一十二乘之，得七百八十，以十三除之，即得。

Now there is a woman washing cups at the riverbank. The ferry official asked, "Why so many cups?" The woman replied, "There are guests at my home." The official asked, "How many guests?" The woman said: "Two people share a meal, three people share soup, four people share meat, and in total sixty-five cups are used. I do not know how many guests there are." Answer: Sixty people.

Problem 22. Wood and Rope

問

今有木，不知長短。引繩度之，餘繩四尺五寸。屈繩量之，不足一尺。問木長幾何?

答

答曰：六尺五寸。

術

術曰：置餘繩四尺五寸，加不足一尺，共五尺五寸。倍之，得一丈一尺。減餘四尺五寸，即得。

Problem 23. Weights of Gold and Silver

問

今有金一斤，令五十人分之。問人得幾何?

答

答曰：人得七銖六絫五黍。

術

術曰：置金一斤，為實。以人五十為法。實如法，即得。

Problem 24. Pulling a Cart

問

今有車運麥，重一千石，日行八十里。問行一百二十里，問運幾何?

答

答曰：一千五百石。

術

術曰：置麥一千石，以一百二十里乘之，得一億二千万。以八十里除之，即得。

Problem 25. Buying Fowl

問

今有共買雞，人出九，盈十一；人出六，不足十六。問人數、雞價各幾何?

答

答曰：人九。雞價七十。

術

術曰：置人出九，盈十一。人出六，不足十六。并盈不足，得二十七。以少減多，餘三。除之，得人數九。以九乘之，減盈十一，得雞價七十。

物不知數 — Problem of the Unknown Quantity

Problem 26, Volume III (卷下第 26 問) — The Chinese Remainder Theorem

問

今有物，不知其數。三三數之賸二，五五數之賸三，七七數之賸二。問物幾何?

答

答曰：二十三。

術

術曰：「三三數之，賸二」，置一百四十；「五五數之，賸三」，置六十三；「七七數之，賸二」，置三十。并之，得二百三十三。以二百一十減之，即得。凡三三數之，賸一，則置七十；五、五數之，賸一，則置二十一；七、七數之，賸一，則置十五。一百六以上，以一百五減之，即得。

Translation:

"Now there is an object, but we do not know its quantity. Counting it in groups of three leaves a remainder of two; counting it in groups of five leaves a remainder of three; counting it in groups of seven leaves a remainder of two. What is the quantity of the object?"

Answer: *Twenty-three.*

Method: *"Counting in groups of three, remainder two," set down one hundred and forty; "Counting in groups of five, remainder three," set down sixty-three; "Counting in groups of seven, remainder two," set down thirty. Add them together to get two hundred and thirty-three. Subtract two hundred and ten from it, and you will obtain the answer. Generally, if counting in groups of three leaves remainder one, set down seventy; if counting in groups of five leaves remainder one, set down twenty-one; if counting in groups of seven leaves remainder one, then set down fifteen. If the total is more than 106, subtract 105 from it; thus you obtain the answer.*

Modern Mathematical Analysis

In modern notation, this problem asks us to find the smallest positive integer N satisfying the system of congruences:

$$\begin{aligned} N &\equiv 2 \pmod{3} \\ N &\equiv 3 \pmod{5} \\ N &\equiv 2 \pmod{7} \end{aligned}$$

The method gives:

$$\begin{aligned} N &= 140 + 63 + 30 - 210 \\ &= 233 - 210 \\ &= 23 \end{aligned}$$

The three key numbers **70**, **21**, and **15** have special properties:

- $70 = 2 \times 5 \times 7 \equiv 1 \pmod{3}$, and $70 \equiv 0 \pmod{5}$, $70 \equiv 0 \pmod{7}$
- $21 = 1 \times 3 \times 7 \equiv 1 \pmod{5}$, and $21 \equiv 0 \pmod{3}$, $21 \equiv 0 \pmod{7}$
- $15 = 1 \times 3 \times 5 \equiv 1 \pmod{7}$, and $15 \equiv 0 \pmod{3}$, $15 \equiv 0 \pmod{5}$

Since $3 \times 5 \times 7 = 105$, the general solution is:

$$N = 70r_1 + 21r_2 + 15r_3 - 105k$$

where r_1, r_2, r_3 are the remainders and k is chosen to make N positive and minimal.

Verification

$$\begin{aligned} 23 \div 3 &= 7 \text{ remainder } \mathbf{2} \quad \checkmark \\ 23 \div 5 &= 4 \text{ remainder } \mathbf{3} \quad \checkmark \\ 23 \div 7 &= 3 \text{ remainder } \mathbf{2} \quad \checkmark \end{aligned}$$

Sunzi's Formula

For general remainders r_1, r_2, r_3:

$$\boxed{N = 70 \times r_1 + 21 \times r_2 + 15 \times r_3 - 105 \times k}$$

where $M = 3 \times 5 \times 7 = 105$ is the least common multiple, and:

- $70 = 105/3 \times 2$ is the multiplier for remainder mod 3
- $21 = 105/5 \times 1$ is the multiplier for remainder mod 5
- $15 = 105/7 \times 1$ is the multiplier for remainder mod 7

The "Sunzi Song" (孫子歌)

A rhymed verse summarizing the method circulated widely among the people and was later recorded by Cheng Dawei in his *Systematic Treatise on Algorithms* (算法統宗, 1592):

> 三人同行七十稀，五樹梅花廿一枝，
> 七子團圓正半月，除百零五便得知。

Translation: "Three people walking together—seventy is rare; five plum trees—twenty-one branches; seven sons reunite—half a full moon (15); subtract one hundred and five, then you will know."

This gives the key numbers: **70** for mod 3, **21** for mod 5, **15** for mod 7, and **105** as the LCM.

Remaining Problems from Volume III

Problem 27. Birds and Beasts

問

今有獸六首四足，禽四首二足。上有七十六首，下有四十六足。問禽、獸各幾何?

答

答曰：八獸，七禽。

術

術曰：倍足以減首，餘，半之，即禽。四乘禽數，減足，餘，半之，即獸。

Now there are beasts with six heads and four feet, and birds with four heads and two feet. Above there are seventy-six heads, below there are forty-six feet. Answer: Eight beasts, seven birds.

Problem 28. Money Exchange

問

今有甲、乙二人，持錢各不知數。甲得乙中半，可滿四十八。乙得甲太半，亦滿四十八。問甲、乙二人元持錢各幾何?

答

答曰：甲持錢三十六。乙持錢二十四。

術

術曰：如方程求之。置二甲、一乙、錢九十六，於右方。置二甲、三乙、錢一百四十四，於左方。以右方二乘左方，上得四，中得六，下得二百八十八錢。以右行再減左行，左上空，中餘四乙，為法；下餘九十六錢，為實。上法，下實，得二十四錢，為乙錢。以減右下九十六，餘七十二，為實；以右上二甲為法。上法、下實，得三十六，為甲錢也。

Problem 29. One Hundred Deer

問

今有百鹿入城，家取一鹿，不盡；又三家共一鹿，適盡。問城中家幾何?

答

答曰：七十五家。

術

術曰：以盈不足取之。假令七十二家，鹿不盡四。令之九十家，鹿不足二十。置七十二於右上，盈四於右下。置九十於左上，不足二十於左下。維乘之，所得，并為實。并盈不足為法。除之，即得。

Now there are one hundred deer entering a city; each household takes one deer, but they do not finish; then three households share one deer and it is just finished. Answer: Seventy-five households.

Problem 30. Three Chickens

問

今有三雞共啄粟一千一粒。雛啄一，母啄二，翁啄四。主責本粟，三雞主各償幾何?

答

答曰：雞雛主一百四十三。雞母主二百八十六。雞翁主五百七十二。

術

術曰：置粟一千一粒，為實。副并三雞所啄粟七粒，為法。除之，得一百四十三粒，為雞雛主所償之數。遞倍之，即得母、翁主所償之數。

Problem 31. Pheasants and Rabbits in a Cage

問

今有雉、兔同籠，上有三十五頭，下九十四足。問雉、兔各幾何?

答

答曰：雉二十三。兔一十二。

術

術曰：上置三十五頭，下置九十四足。半其足，得四十七。以少減多，再命之，上三除下三，上五除下五。下有一除上一，下有二除上二，即得。

This is the classic "chicken and rabbit in a cage" (雞兔同籠) problem.

Alternative method: 又術曰：上置頭，下置足。半其足，以頭除足，以足除頭，即得。

Problem 32. Fish in a Ditch

問

今有九里渠，三寸魚，頭頭相次。問魚得幾何?

答

答曰：五萬四千。

術

術曰：置九里，以三百步乘之，得二千七百步。又以六尺乘之，得一萬六千二百尺。上十之，得一十六萬二千寸。以魚三寸除之，即得。

Problem 33. Wheel Turns from Luoyang to Chang'an

問

今有長安、洛陽相去九百里。車輪一匝一丈八尺。欲自洛陽至長安，問輪匝幾何?

答

答曰：九萬匝。

術

術曰：置九百里，以三百步乘之，得二十七萬步。又以六尺乘之，得一百六十二萬尺。以車輪一丈八尺為法。除之，即得。

Problem 34. Nine Dikes

問

今有出門望見九隄。隄有九木，木有九枝，枝有九巢，巢有九禽，禽有九雛，雛有九毛，毛有九色。問各幾何?

答

答曰：木八十一。枝七百二十九。巢六千五百六十一。禽五萬九千四十九。雛五十三萬一千四百四十一。毛四百七十八萬二千九百六十九。色四千三百四萬六千七百二十一。

術

術曰：置九隄，以九乘之，得木之數。又以九乘之，得枝之數。又以九乘之，得巢之數。又以九乘之，得禽之數。又以九乘之，得雛之數。又以九乘之，得毛之數。又以九乘之，得色之數。

Item	Count
Trees (木)	$9^2 = 81$
Branches (枝)	$9^3 = 729$
Nests (巢)	$9^4 = 6{,}561$
Birds (禽)	$9^5 = 59{,}049$
Chicks (雛)	$9^6 = 531{,}441$
Feathers (毛)	$9^7 = 4{,}782{,}969$
Colors (色)	$9^8 = 43{,}046{,}721$

Table 1: The powers of nine from the Nine Dikes problem

Problem 35. Three Daughters

問

今有三女，長女五日一歸，中女四日一歸，少女三日一歸。問三女幾何日相會?

答

答曰：六十日。

術

術曰：置長女五日、中女四日、少女三日，於右方。各列一算於左方。維乘之，各得所到數：長女十二到，中女十五到，少女二十到。又各以歸日乘到數，即得。

Now there are three daughters: the eldest returns home every five days, the middle every four days, and the youngest every three days. Ask: After how many days will they meet again? Answer: Sixty days. This is a least common multiple problem: $lcm(3, 4, 5) = 60$.

Problem 36. Predicting the Gender of an Unborn Child

問

今有孕婦行年二十九，難九月。未知所生?

答

答曰：生男。

術

術曰：置四十九，加難月，減行年。所餘，以天除一，地除二，人除三，四時除四，五行除五，六律除六，七星除七，八風除八，九州除九。其不盡者，奇則為男，耦則為女。

Now there is a pregnant woman whose age is twenty-nine, giving birth in the ninth month. It is unknown what will be born? Answer: A boy will be born. Method: Set forty-nine, add the month of childbirth, subtract the age. The remainder is divided by: Heaven subtracts one, Earth subtracts two, Humanity subtracts three, Four Seasons subtract four, Five Elements subtract five, Six Laws subtract six, Seven Stars subtract seven, Eight Winds subtract eight, Nine Provinces subtract nine. If the remainder is not exhausted, an odd number means a boy will be born, and an even number means a girl will be born.

Epilogue

The *Sunzi Suanjing* stands as a remarkable testament to the sophistication of ancient Chinese mathematics. While the text covers basic arithmetic—units of measurement, multiplication tables, and simple fractions—its greatest legacy lies in the brief but brilliant "Problem of the Unknown Quantity" that opens the door to modern number theory.

The Chinese Remainder Theorem, as this method came to be known in the West, would not be fully formalized in Europe for over a millennium. The algorithm described by Sunzi contains the essential ingredients of the modern theorem: the construction of auxiliary numbers with specific divisibility properties, their combination with the given remainders, and reduction modulo the least common multiple.

The influence of this text extended far beyond China. The problem circulated as a mathematical puzzle throughout East Asia, and the general method developed by Qin Jiushao represents one of the great achievements of medieval mathematics anywhere in the world.

In Europe, similar methods were not developed until the work of Euler (1743) and Gauss (1801). In 1852, the British missionary Alexander Wylie published notes on Chinese science, introducing this problem to European mathematicians. In 1876, the German mathematician L. Matthiessen first pointed out that Sunzi's solution was consistent with Gauss's method.

Typeset from the classical Chinese text as preserved in the
Chinese Text Project (https://ctext.org/sunzi-suan-jing)